

A* Search Algorithm

PROGRAM:

```
import heapq

def a_star(graph, start, goal, heuristic):
    open_list = []
    heapq.heappush(open_list, (0, start))
    g_cost = {node: float('inf') for node in graph}
    g_cost[start] = 0
    parent = {start: None}

    while open_list:
        _, current = heapq.heappop(open_list)

        if current == goal:
            path = []
            while current is not None:
                path.append(current)
                current = parent[current]
            return path[::-1]

        for neighbor, cost in graph[current]:
            new_cost = g_cost[current] + cost
            if new_cost < g_cost[neighbor]:
                g_cost[neighbor] = new_cost
                f_cost = new_cost + heuristic[neighbor]
                heapq.heappush(open_list, (f_cost, neighbor))
                parent[neighbor] = current

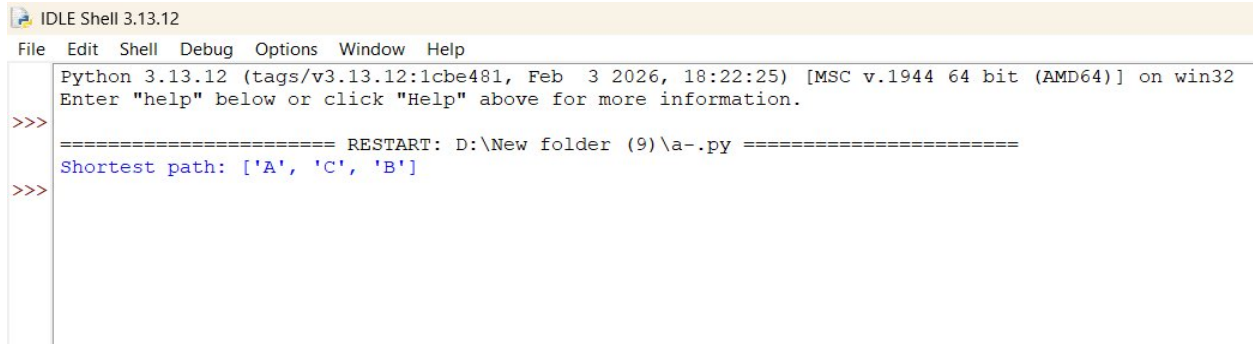
    return None

graph = {
    'A': [('C', 1), ('B', 5)],
    'C': [('B', 1)],
    'B': []
}

heuristic = {
    'A': 2,
    'C': 1,
    'B': 0
}
```

```
result = a_star(graph, 'A', 'B', heuristic)
print("Shortest path:", result)
```

OUTPUT:



The screenshot shows an IDLE Shell window titled "IDLE Shell 3.13.12". The menu bar includes "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The shell displays the following text:

```
Python 3.13.12 (tags/v3.13.12:1cbe481, Feb  3 2026, 18:22:25) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:\New folder (9)\a-.py =====
Shortest path: ['A', 'C', 'B']
>>>
```